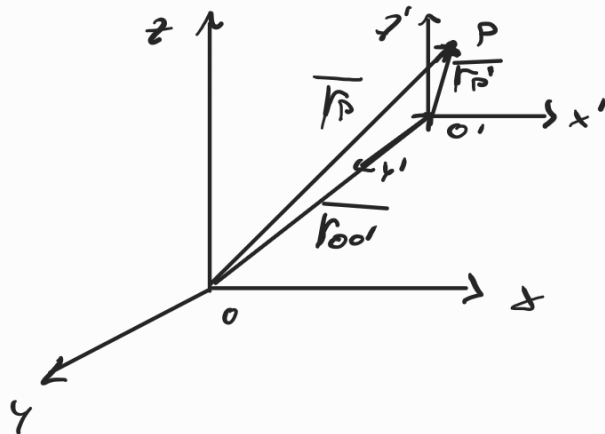


MOTO RELATIVO



$$\overline{r_P} = \overline{r_{OO'}} + \overline{r_{P'}}$$

$$\overline{r_{P'}} = x \hat{i} + y \hat{j} + z \hat{k}$$

$$\overline{v} = \frac{d\overline{r_P}}{dt} = \underbrace{\dot{x} \hat{i} + \dot{y} \hat{j} + \dot{z} \hat{k}}_{v_{rel}} + \underbrace{x \dot{\hat{i}} + y \dot{\hat{j}} + z \dot{\hat{k}}}_{v_{tr}} + \underbrace{\frac{d\overline{r_{OO'}}}{dt}}_{v_T}$$

$$\overline{a} = \frac{d\overline{v}}{dt} = \underbrace{\ddot{x} \hat{i} + \ddot{y} \hat{j} + \ddot{z} \hat{k}}_{a_{rel}} + \underbrace{x \ddot{\hat{i}} + y \ddot{\hat{j}} + z \ddot{\hat{k}}}_{a_{tr}} + \underbrace{\frac{d^2 \overline{r_{OO'}}}{dt^2}}_{a_T} + \underbrace{2 \left(\dot{x} \dot{\hat{i}} + \dot{y} \dot{\hat{j}} + \dot{z} \dot{\hat{k}} \right)}_{a_c}$$

a_c = TERMINE DI ACCELERAZIONE DOVUTO ALLE FORZE APPARENTI DEL SISTEMA DI RIFERIMENTO NON INERZIALE

TEOREMA DELL'IMPULSO

$$\begin{aligned} \overline{I} &= \Delta \overline{p} \\ \overline{I} &= \int_t^{\Delta t + t} \overline{F} dt = \int_t^{\Delta t + t} m \frac{d\overline{v}}{dt} dt = \int_t^{\Delta t + t} d\overline{p} = \overline{p}(\Delta t + t) - \overline{p}(\Delta t) = \\ &= \overline{p}_f - \overline{p}_i = \Delta \overline{p} \end{aligned}$$

MOLE IN SERIE

$$k_2 x_2 = F = k_1 x_1 \quad X = x_1 + x_2 = \frac{F}{k_1} + \frac{F}{k_2} = F \left(\frac{1}{k_1} + \frac{1}{k_2} \right)$$

$$X = \frac{F}{k} \quad \frac{1}{k} = \frac{1}{k_1} + \dots + \frac{1}{k_n}$$

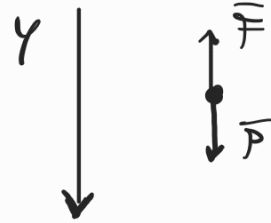
FORZA IN PARALLELO

$$F = F_{H1} + F_{H2} = k_1 \Delta x + k_2 \Delta x = \Delta x (k_1 + k_2)$$

$$F = k \Delta x \quad k = k_1 + \dots + k_n$$

APPARATO VISCOZO

$$\vec{F} = -b\vec{v} \quad b = mk$$



$$mg - b\vec{v} = m\vec{a}$$

$$mg - kv = m\vec{a}$$

$$g - kv = a = \frac{dv}{dt}$$

$$dt = \frac{dv}{g - kv} \Rightarrow \int_0^t dt = \int_0^v \frac{dv}{g - kv}$$

$$\Rightarrow t = -\frac{1}{k} \ln(g - kv) - \ln(g) = -\frac{1}{k} \ln\left(\frac{g - kv}{g}\right)$$

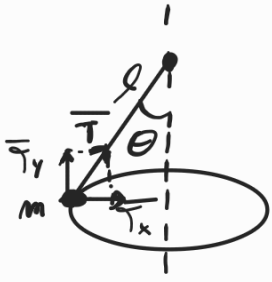
$$\ln\left(\frac{g - kv}{g}\right) = -kt \quad \frac{g - kv}{g} = e^{-kt}$$

$$v(t) = \left(1 - e^{-kt}\right) \frac{g}{k}$$

$$v(kt=1) \approx \frac{2}{3} \frac{g}{k} \quad \tau = \frac{1}{k}$$

$$v(t \rightarrow \infty) = \frac{g}{k} \text{ costante}$$

PERÍODO CONICO



$$T_x = m a_c = m \frac{v^2}{R} = m \omega^2 R$$

$$R = l \sin \theta \quad T_x = m \omega^2 l \sin \theta$$

$$T \sin \theta = m \omega^2 l \sin \theta \quad T = m \omega^2 l$$

$$T_y = mg \quad T \cos \theta = mg \quad T = \frac{mg}{\cos \theta}$$

$$T = m \omega^2 l = \frac{mg}{\cos \theta} \Rightarrow \omega^2 = \frac{g}{l \cos \theta}$$

$$\omega = \frac{2\pi}{T} \quad T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{l}{g \cos \theta}}$$

PERÍODO MATEMÁTICO

DINAMICA DEI SISTEMI DI PUNTI MATERIALI:

$$\overline{F}_S^{\text{EST}} + \sum_{\substack{i=1 \\ i \neq S}}^N \overline{F}_{Si} = m_S \frac{d\overline{V}_S}{dt} = \frac{d m_S \overline{V}_S}{dt} = \frac{d \overline{P}_S}{dt}$$

PRINCIPIO DI SOVRAPPOSIZIONE DEGLI EFFETTI

CORPO ESTERNO:

$$\sum_{S=1}^N \overline{F}_S^{\text{EST}} + \sum_{S=1}^N \sum_{\substack{i=1 \\ i \neq S}}^N \overline{F}_{Si} = \sum_{S=1}^N \frac{d \overline{P}_S}{dt}$$

$\overline{F}_{is} = -\overline{F}_{Si}$

$$\Rightarrow \underbrace{\sum_{S=1}^N \overline{F}_S^{\text{EST}}}_{\overline{F}_{\text{TOT}}^{\text{EST}}} = \sum_{S=1}^N \frac{d \overline{P}_S}{dt} = \frac{d}{dt} \underbrace{\sum_{S=1}^N \overline{P}_S}_{\overline{P}_{\text{TOT}}}$$

$$\Rightarrow \boxed{\overline{F}_{\text{TOT}}^{\text{EST}} = \frac{d \overline{P}_{\text{TOT}}}{dt}}$$

I EQUAZIONE
CARDINALE

$$\rho = \lim_{\Delta V \rightarrow 0} \frac{\Delta m}{\Delta V} = \frac{dm}{dV}$$

$$dm = \rho dV$$

$$\overline{P}_{\text{TOT}} = \int_V \overline{v}(\vec{r}) dm \Rightarrow \underline{\underline{\overline{P}_{\text{TOT}} = \int_V \overline{v}(\vec{r}) \rho(\vec{r}) dV}}$$

$$\overline{F}_{\text{TOT}}^{\text{EXT}} = 0 \Rightarrow \frac{d \overline{P}_{\text{TOT}}}{dt} = 0 \Rightarrow \overline{P}_{\text{TOT}} \text{ COSTANTE}$$

SISTEMA ISOLATO $\Rightarrow$ CONSERVAZIONE DELLA QUANTITÀ DI MOTO

$$\overline{r_{cm}} = \frac{\sum_i^N m_i \overline{r_i}}{\sum_i^N m_i} = \frac{1}{M} \sum_i^N m_i \overline{r_i}$$

$$M = \int_V \rho(\vec{r}) dV \quad \overline{r_{cm}} = \frac{1}{M} \int_V \vec{r} \rho(\vec{r}) dV$$

$$\overline{v_{cm}} = \frac{d\overline{r_{cm}}}{dt} = \frac{1}{M} \sum_i^N m_i \frac{d\overline{r_i}}{dt} = \frac{1}{M} \sum_i^N m_i \overline{v_i} = \frac{\overline{p_{tot}}}{M}$$

$$\boxed{\overline{p_{tot}} = M \overline{v_{cm}}}$$

$$\Rightarrow \overline{F_{tot}^{ext}} = \frac{d\overline{p_{tot}}}{dt} = M \frac{d\overline{v_{cm}}}{dt} = M \overline{a_{cm}}$$

$$\boxed{\overline{F_{tot}^{ext}} = M \overline{a_{cm}}}$$

TEOREMA DEL CENTRO
DI MASSA

MOMENTO DI UNA FORZA RISPETTO A UN POLO O :

$$\overline{\tau_o} = \overline{r_o} \times \overline{F}$$

INFORMAZIONI CONTENUTE IN $\overline{\tau_o}$:

$$\overline{\tau_{tot}} = \sum_{s=1}^N \overline{r_{so}} \times \overline{F_s}$$

PUNTO E DIREZIONE DI APPLICAZIONE
DI UNA FORZA $\overline{F}$

MOMENTO FORZA PESO :

$$\overline{\tau_p} = \sum_{s=1}^N \overline{r_{so}} \times m_s \overline{g} = \left(\sum_s \overline{r_{so}} m_s \right) \times \overline{g} = \left(\frac{1}{M} \sum_s \overline{r_{so}} m_s \right) \times M \overline{g}$$

$$\Rightarrow \overline{\tau_p} = \overline{r_{Ho}} \times \overline{F_p}$$

CONSIDERIAMO UN SISTEMA DI N PARTICELLE:

$$\overline{r_{s0}} \times \overline{F_s^{EXT}} + \sum_{\substack{i=1 \\ i \neq s}}^N \overline{r_{s0}} \times \overline{F_{si}} = \overline{r_{s0}} \times \frac{d\overline{p_s}}{dt}$$

$$\Rightarrow \sum_s \overline{r_{s0}} \times \overline{F_s^{EXT}} + \underbrace{\sum_s \sum_{i \neq s} \overline{r_{s0}} \times \overline{F_{si}}}_{\overline{r_{i0}} \times \overline{F_{is}} = -\overline{r_{j0}} \times \overline{F_{ji}}} = \sum_s \overline{r_{s0}} \times \frac{d\overline{p_s}}{dt}$$

$$\Rightarrow \sum_s \overline{r_{s0}} \times \overline{F_s^{EXT}} = \sum_s \overline{r_{s0}} \times \frac{d\overline{p_s}}{dt}$$

$$\Rightarrow \frac{d(\overline{r_{s0}} \times \overline{p_s})}{dt} = \overline{r_{s0}} \times \frac{d\overline{p_s}}{dt} + \underbrace{\frac{d\overline{r_{s0}}}{dt} \times \overline{p_s}}_{\overline{v_s} \parallel \overline{p_s}} = \overline{r_{s0}} \times \frac{d\overline{p_s}}{dt}$$

$$\Rightarrow \sum_s \overline{r_{s0}} \times \overline{F_s^{EXT}} = \sum_s \frac{d}{dt} (\overline{r_{s0}} \times \overline{p_s}) \Rightarrow \overline{L_{s0}} = \overline{r_{s0}} \times \overline{p_s}$$

$$\boxed{\overline{\tau_{TOT}^{EXT}} = \frac{d\overline{L_{TOT}}}{dt}}$$

II EQUAZIONE CARDINALE

$$\overline{\tau_{TOT}^{EXT}} = \sum_s \overline{r_{s0}} \times \overline{F_s^{EXT}}$$

$$\overline{L_{TOT}} = \sum_s \overline{L_{s0}} = \sum_s \overline{r_{s0}} \times \overline{p_s}$$

$$\overline{L_{TOT}} = \int_V \overline{r_0} \times \overline{v(r)} \rho(r) dV$$

LA DERIVATA TEMPORALE DEL MOMENTO ANGOLARE È UGUALE AL MOMENTO RISULTANTE DELLE FORZE ESTERNE.

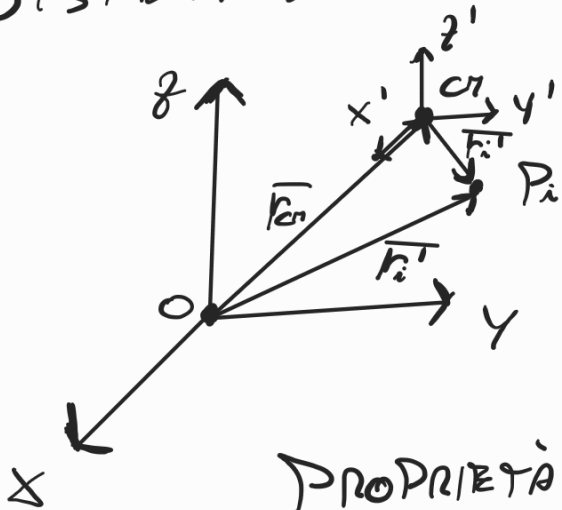
LA VARIAZIONE TEMPORALE DEL MOMENTO ANGOLARE TOTALE RISPETTO A UN POLO DIPENDE SOLO DAL MOMENTO RISULTANTE DELLE FORZE ESTERNE NELLO STESSO POLO.

$$\boxed{\frac{d\overline{L_{TOT}}}{dt} = -\overline{v_0} \times \overline{p_{TOT}} + \overline{\tau_{TOT}^{EXT}}}$$

$$\overline{L_{\text{tot}}^{\text{EXT}}} = \frac{d\overline{L_{\text{tot}}}}{dt} = 0 \Rightarrow \overline{L_{\text{tot}}} \text{ COSTANTE}$$

PRINCIPIO DI CONSERVAZIONE DEL MOMENTO ANGOLARE

SISTEMA DI RIFERIMENTO CENTRO DI MASSA



$$S \parallel S'$$

$$\vec{r}_i = \vec{r}_i' + \vec{r}_{cm}$$

$$\frac{d\vec{r}_i}{dt} = \vec{v}_i = \vec{v}_i' + \vec{v}_{cm}$$

PROPRIETÀ S' (SISTEMA CENTRO DI MASSA)

I) $\overline{r_{cm}'} = 0$ DIM:

$$\begin{aligned} \vec{r}_i' &= \vec{r}_i - \vec{r}_{cm} \Rightarrow \sum_i m_i \vec{r}_i' = \sum_i m_i (\vec{r}_i - \vec{r}_{cm}) = \\ &= \underbrace{\sum_i m_i \vec{r}_i}_{M \vec{r}_{cm}} - \underbrace{\sum_i m_i}_{M} \vec{r}_{cm} = 0 \end{aligned}$$

II) $\overline{v_{cm}'} = 0$ DIM:

$$M \overline{v_{cm}'} = \sum_i m_i \overline{v_i'} = \sum_i m_i (\overline{v_i} - \overline{v_{cm}}) = \sum_i m_i \overline{v_i} - M \overline{v_{cm}} = 0$$

III) $\sum_i m_i \vec{a}_i' = 0$ DIM:

$$\begin{aligned} \frac{d\overline{p_{cm}}}{dt} &= \overline{F_{\text{EXT}}} \quad \overline{F_i^{\text{EXT}}} + \overline{F_i^{\text{INT}}} - m_i \overline{a_{cm}} = m_i \overline{a_i'} \\ &\Rightarrow \sum_i \left(\overline{F_i^{\text{EXT}}} + \overline{F_i^{\text{INT}}} - m_i \overline{a_{cm}} \right) = \sum_i m_i \overline{a_i'} \\ &\Rightarrow 0 = \overline{F_{\text{TOT}}^{\text{EXT}}} - M \overline{a_{cm}} = \sum_i m_i \overline{a_i'} \end{aligned}$$

$$\text{IV) } \frac{d\bar{\mathcal{L}}}{dt} = \overline{\mathcal{H}_{\text{EXT}}} \quad \text{Dim:}$$

$$\bar{\ell}_i = \bar{\mathbf{r}}_i \times m_i \bar{\mathbf{v}}_i \Rightarrow \sum_i \bar{\ell}_i = \bar{\mathcal{L}} = \sum_i \bar{\mathbf{r}}_i \times m_i \bar{\mathbf{v}}_i$$

$$\begin{aligned} \frac{d\bar{\mathcal{L}}}{dt} &= \frac{d}{dt} \left(\sum_i \bar{\mathbf{r}}_i \times m_i \bar{\mathbf{v}}_i \right) = \sum_{i=1}^N \left(\underbrace{\frac{d\bar{\mathbf{r}}_i}{dt}}_{\bar{\mathbf{v}}_i} \times m_i \bar{\mathbf{v}}_i \right) + \\ &+ \sum_i \left(\bar{\mathbf{r}}_i \times m_i \underbrace{\frac{d\bar{\mathbf{v}}_i}{dt}}_{\substack{\bar{\mathbf{F}}_i^{\text{EXT}} + \bar{\mathbf{F}}_i^{\text{INT}}}} \right) = \sum_i \bar{\mathbf{r}}_i \times \bar{\mathbf{F}}_i^{\text{EXT}} = \overline{\mathcal{H}_{\text{EXT}}} \end{aligned}$$

TEOREMA DELLE FORZE VIVE

$$\delta W_i = \bar{\mathbf{F}}_i \circ d\bar{\mathbf{r}}_i = (\bar{\mathbf{F}}_i^{\text{EXT}} + \bar{\mathbf{F}}_i^{\text{INT}}) \circ d\bar{\mathbf{r}}_i$$

$$dW = \sum_i \delta W_i = \sum_i \bar{\mathbf{F}}_i^{\text{EXT}} \circ d\bar{\mathbf{r}}_i + \sum_i \bar{\mathbf{F}}_i^{\text{INT}} \circ d\bar{\mathbf{r}}_i$$

$$W = \int_{\mathcal{Q}} \delta W = \int_{\mathcal{Q}} \sum_i \delta W_i$$

$$\begin{aligned} \sum_i \bar{\mathbf{F}}_i^{\text{INT}} \circ d\bar{\mathbf{r}}_i &= (\dots) + \bar{\mathbf{F}}_{Si} \circ d\bar{\mathbf{r}}_i + (\dots) + \bar{\mathbf{F}}_{iS} \circ d\bar{\mathbf{r}}_i + (\dots) = \\ &\quad \parallel \\ &\quad -\bar{\mathbf{F}}_{Si} \\ &= (\dots) + \bar{\mathbf{F}}_{Si} \circ (d\bar{\mathbf{r}}_i - d\bar{\mathbf{r}}_S) + (\dots) \end{aligned}$$

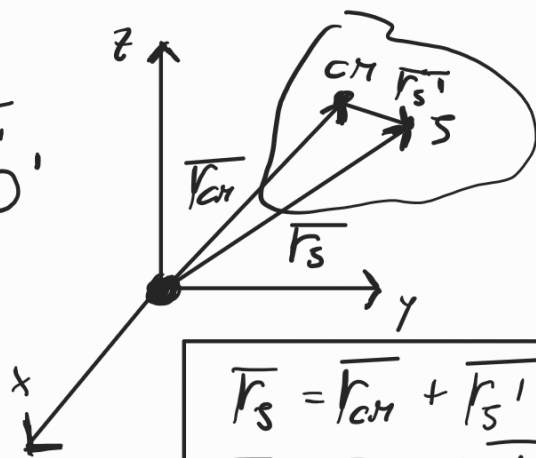
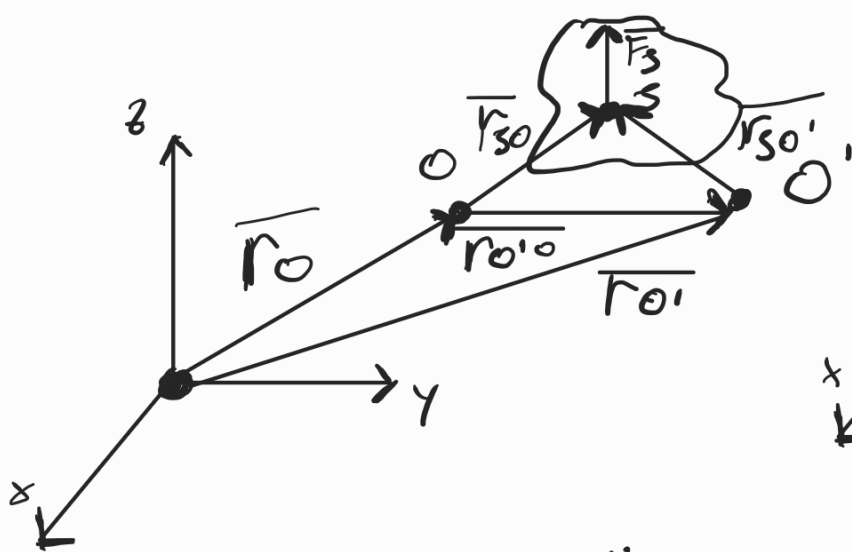
$$W = W_{\text{EXT}} + W_{\text{INT}} = \Delta K \quad W_{\text{EXT}} = 0$$

$$\delta W = \sum_i \bar{\mathbf{F}}_i \circ d\bar{\mathbf{r}}_i = m_i \bar{\mathbf{a}}_i \circ d\bar{\mathbf{r}}_i = m_i \frac{d\bar{\mathbf{v}}_i}{dt} \circ d\bar{\mathbf{r}}_i =$$

$$= m_i \frac{d\bar{\mathbf{v}}_i}{dt} \circ \bar{\mathbf{v}}_i dt = \frac{1}{2} m_i d(v_i^2) = \delta W$$

$$W = \int \sum_i \frac{1}{2} m_i d(v_i^2) = \sum_i \int \frac{1}{2} m_i d(v_i^2)$$

$$\Rightarrow W = \sum_i \Delta K_i = \Delta K = K_F - K_i$$



$$\begin{aligned}\vec{r}_S &= \vec{r}_{cm} + \vec{r}_{S'} \\ \vec{v}_S &= \vec{v}_{cm} + \vec{v}_{S'}\end{aligned}$$

$$\begin{aligned}\vec{L}_{rot}^{ext'} &= \sum_S \vec{r}_{SO'} \times \vec{F}_S^{ext} = \sum_S (\vec{r}_{SO} - \vec{r}_{O'O}) \times \vec{F}_S^{ext} = \\ &= \sum_S \vec{r}_{SO} \times \vec{F}_S^{ext} - \vec{r}_{O'O} \times \sum_S \vec{F}_S^{ext}\end{aligned}$$

$$\vec{L}_{rot}^{ext'} = \vec{L}_{rot}^{ext} - \vec{r}_{O'O} \times \vec{F}_{tot}^{ext}$$

I TEOREMA DI
KÖNIG

$$\vec{L} = \vec{L}' + \underbrace{\vec{r}_{cm} \times \vec{p}_{cm}}_{\vec{L}_{cm}}$$

$$\sum_S m_S \vec{v}_{S'} = \sum_S m_S \frac{d\vec{r}_{S'}}{dt} = \frac{d}{dt} \underbrace{\sum_S m_S \vec{r}_S}_{M \vec{v}_{cm}} - \underbrace{\sum_S m_S \vec{v}_{cm}}_{M \vec{v}_{cm}} = \vec{0}$$

$$\begin{aligned}E_c^{tot} &= \sum_S \left(\frac{1}{2} m_S v_S^2 \right) = \sum_S \left[\frac{1}{2} m_S (\vec{v}_{cm} + \vec{v}_{S'})^2 \right] = \\ &= \sum_S \frac{1}{2} m_S v_{cm}^2 + \sum_S \frac{1}{2} m_S v_{S'}^2 + \vec{v}_{cm} \cdot \sum_S m_S \vec{v}_{S'}\end{aligned}$$

$$E_c^{tot} = \underbrace{\frac{1}{2} M v_{cm}^2}_{E_{cm}^{tot}} + \underbrace{\sum_S \frac{1}{2} m_S v_{S'}^2}_{E_{int}^{tot}}$$

$$E_c^{tot} = E_{cm}^{tot} + E_{int}^{tot}$$

II TEOREMA DI
KÖNIG

L'ENERGIA CINETICA TOTALE DI UN SISTEMA DI PARTICELLE
 È LA SOMMA DELL'ENERGIA CINETICA INTERNA E
 DELL'ENERGIA CINETICA $\frac{1}{2} M v_{cm}^2$ CHE IL SISTEMA AVREBBE
 SE FOSSE UN PUNTO MATERIALE NEL CENTRO DI MASSA

PER $E_{c}^{INT} \ll E_{cm}^{TOT}$

$$E_c^{TOT} \approx E_{cm}$$

APPROSSIMAZIONE
 A PUNTO MATERIALE

CONSERVATIONE DELL'ENERGIA IN UN SISTEMA ISOLATO

$$\delta W_S = \overline{F_S^{\text{EXT}}} \circ d\overline{r}_S + \sum_{i \neq S} \overline{F}_{Si} \circ d\overline{r}_S$$

$$\delta W_S = \overline{F_S^{\text{EXT}}} \circ d\overline{r}_S + \sum_{i \neq S} \overline{F}_{Si} \circ d\overline{r}_S = dE_{cS} = d\left(\frac{1}{2} m_S v_S^2\right)$$

$$\delta W = \sum_S \delta W_S = \sum_S \overline{F_S^{\text{EXT}}} \circ d\overline{r}_S + \sum_S \sum_{i \neq S} \overline{F}_{Si} \circ d\overline{r}_S =$$

$$\delta W = d \sum_S E_{cS} = dE_c^{\text{TOT}}$$

- COPPIA DI PARTICELLE i e S

$$\delta W_{Si} = \overline{F}_{Si} \circ d\overline{r}_S + \overline{F}_{iS} \circ d\overline{r}_i = \overline{F}_{Si} \circ (d\overline{r}_S - d\overline{r}_i)$$

$$\delta W_{Si} = \overline{F}_{Si} \circ d\overline{r}_{Si} \stackrel{\substack{\parallel \\ \text{2 PARTICELLE} \\ \text{TUTTE FORZE CONSERVATIVE MICROSCOPICAMENTE}}}{=} \stackrel{\substack{\parallel \\ -\overline{F}_{Si} \\ \parallel}}{=} -dE_P(\overline{r}_{Si})$$

$$\Rightarrow \delta W = \sum_S \overline{F_S^{\text{EXT}}} \circ d\overline{r}_S - \frac{1}{2} \sum_S \sum_{i \neq S} E_P(\overline{r}_{Si}) \Big] dE_P^{\text{INT}}$$

$$\Rightarrow \delta W = \sum_S \overline{F_S^{\text{EXT}}} \circ d\overline{r}_S - dE_P^{\text{INT}} \stackrel{\text{II KÖNIG}}{=} d\left(\frac{1}{2} M v_{CM}^2\right) + dE_c^{\text{INT}}$$

$$\Rightarrow \sum_S \overline{F_S^{\text{EXT}}} \circ d\overline{r}_S = d\left(\frac{1}{2} M v_{CM}^2\right) + dE_{\text{TOT}}^{\text{INT}}$$

$$E_{\text{TOT}}^{\text{INT}} = E_P^{\text{INT}} + E_c^{\text{INT}} = \frac{1}{2} \sum_S \sum_{i \neq S} E_P(\overline{r}_{Si}) + \frac{1}{2} \sum_S m_S v_S^2$$

SISTEMA ISOLATO $\Rightarrow \overline{F_S^{\text{EXT}}} = 0 \Rightarrow v_{CM}$ COSTANTE

$$\Rightarrow d\left(\frac{1}{2} M v_{CM}^2\right) = 0 \Rightarrow dE_{\text{TOT}}^{\text{INT}} = 0 \Rightarrow E_{\text{TOT}}^{\text{INT}} \text{ COSTANTE}$$

FORZA IMPULSIVA: INTERVALLO BREVE, INTENSITÀ MAGGIORE
URTO (EVENTO IMPULSIVO): DUE CORPI INTERAGENTI CON FORZE IMPULSIVE

TEOREMA DELL'IMPULSO:

$$\overline{I}(t_1, t_2) = \int_{t_1}^{t_2} \overline{F}(t) dt \quad \text{IMPULSO}$$

$$\overline{I} = \int_{t_1}^{t_2} \frac{d\overline{P}}{dt} dt = \int_{\overline{P}(t_1)}^{\overline{P}(t_2)} d\overline{P} = \overline{P}(t_2) - \overline{P}(t_1) = \Delta \overline{P}$$

L'IMPULSO DELLA FORZA RESULTANTE APPLICATA IN UN CORPO ΔT
 È PARI ALLA VARIAZIONE DI QUANTITÀ DI MOTO $\Delta \overline{P}$

URTO = FORZA DI INTERAZIONE INTERNA AL SISTEMA

$\Rightarrow$ CONSERVAZIONE DI MOTO DEL SISTEMA

$$m_1 \overline{v_{1i}} + m_2 \overline{v_{2i}} = m_1 \overline{v_{1f}} + m_2 \overline{v_{2f}}$$

CIASCUN PUNTO SUBISCE UNA VARIAZIONE DI MOTO

$$\overline{I}_1 = \int_{t_i}^{t_f} \overline{F}_{12} dt = m_1 \overline{v_{1f}} - m_1 \overline{v_{1i}} = \overline{P_{1f}} - \overline{P_{1i}}$$

$$\overline{I}_2 = \int_{t_i}^{t_f} \overline{F}_{21} dt = m_2 \overline{v_{2f}} - m_2 \overline{v_{2i}} = \overline{P_{2f}} - \overline{P_{2i}}$$

APPROSSIMAZIONE IMPULSIVA:

$$\overline{I}_1 = \int_{t_i}^{t_f} (\overline{F}_{12} + \overline{F}_1^{\text{EXT}}) dt \approx \int_{t_i}^{t_f} \overline{F}_{12} dt$$

$$\int_{t_i}^{t_f} \overline{F}_1^{\text{EXT}} dt \ll \int_{t_i}^{t_f} \overline{F}_{12} dt$$

$$\overline{P}_i = \overline{P}_f = \overline{P}_{\text{TOT}} = M \overline{v_{\text{CM}}}$$

IN ASSENZA DI FORZE ESTERNE IMPULSIVE $\Delta \overline{P}_{\text{TOT}} = 0$

URTO ELASTICO:

CONSERVAZIONE DELL'ENERGIA CINETICA TOTALE DEL SISTEMA

$$m_1 \overline{V_{1i}} + m_2 \overline{V_{2i}} = m_1 \overline{V_{1f}} + m_2 \overline{V_{2f}}$$

CONSERVAZIONE
QUANTITÀ DI MOTO

$$\frac{1}{2} m_1 V_{1i}^2 + \frac{1}{2} m_2 V_{2i}^2 = \frac{1}{2} m_1 V_{1f}^2 + \frac{1}{2} m_2 V_{2f}^2$$

CONSERVAZIONE ENERGIA CINETICA

$$\Rightarrow V_{1f} = \frac{2 m_2 V_{2i} + (m_1 - m_2) V_{1i}}{m_1 + m_2}$$
$$\Rightarrow V_{2f} = \frac{2 m_1 V_{1i} + (m_2 - m_1) V_{2i}}{m_1 + m_2}$$

SE $m_1 = m_2$

$$\Rightarrow V_{1f} = V_{2i}$$

$$V_{2f} = V_{1i}$$

SE $V_{2i} = 0$

$$\Rightarrow V_{1f} = \frac{(m_1 - m_2) V_{1i}}{m_1 + m_2}$$

$$V_{2f} = \frac{2 m_1 V_{1i}}{m_1 + m_2}$$

SE $m_1 \gg m_2$ $V_{2i} = 0$

$$\Rightarrow V_{1f} \approx V_{1i}$$

$$V_{2f} \approx 2 V_{1i}$$

SE $m_1 \ll m_2$ $V_{2i} = 0$

$$\Rightarrow V_{1f} \approx -V_{1i}$$

$$V_{2f} \approx 0$$

URTO PERFETTAMENTE ANELASTICO:

I DUE CORPI RISULTANO UNITI DOPO L'URTO

$$V_{1f} = V_{2f} = V_f = \text{CONSERVAZIONE QUANTITÀ DI MOTO} \Rightarrow$$

$$\Rightarrow V_f = \frac{m_1 V_{1i} + m_2 V_{2i}}{m_1 + m_2}$$

CONSERVAZIONE $\overline{P}$ $\checkmark$
CONSERVAZIONE E_k $\times$

NEL SISTEMA DI RIFERIMENTO CENTRO DI MASSA LA QUANTITÀ DI MOTO TOTALE È NULLA

$$\overline{V}_{1(i)} = \overline{V}_{CM} + \overline{V}'_{1(i)}$$

$$\Rightarrow \cancel{(m_1 + m_2)} \overline{V}_{CM} = m_1 \overline{V}'_{1(f)} + m_2 \overline{V}'_{2(f)} + \cancel{(m_1 + m_2)} \overline{V}_{CM}$$

$$\Rightarrow \begin{cases} m_1 \overline{V}'_{1i} + m_2 \overline{V}'_{2i} = \overline{P}'_{1i} + \overline{P}'_{2i} = 0 \Rightarrow \overline{P}'_{1i} = -\overline{P}'_{2i} \\ m_1 \overline{V}'_{1f} + m_2 \overline{V}'_{2f} = \overline{P}'_{1f} + \overline{P}'_{2f} = 0 \Rightarrow \overline{P}'_{1f} = -\overline{P}'_{2f} \end{cases}$$

CONSERVAZIONE DELL'ENERGIA CINETICA NEL SISTEMA CM:

$$\frac{1}{2} m_1 V_{1i}'^2 + \frac{1}{2} m_2 V_{2i}'^2 = \frac{1}{2} m_1 V_{1f}'^2 + \frac{1}{2} m_2 V_{2f}'^2$$

$$P = mV \quad \left(\frac{P_{1i}'^2}{m_1} + \frac{P_{2i}'^2}{m_2} = \frac{P_{1f}'^2}{m_1} + \frac{P_{2f}'^2}{m_2} \right)$$

$$\Rightarrow P_{1(i)i}^2 = P_{1(i)f}^2 \longrightarrow |\overline{V}'_{1(i)i}| = |\overline{V}'_{1(i)f}|$$

$$\text{COEFFICIENTE DI RESTITUZIONE } e = \frac{|\overline{V}'_{1f}|}{|\overline{V}'_{1i}|}$$

$$\Delta E_c = E_{cf}' - E_{ci}' = \left(\frac{1}{2} m_1 V_{1f}'^2 + \frac{1}{2} m_2 V_{2f}'^2 \right) - \left(\frac{1}{2} m_1 V_{1i}'^2 + \frac{1}{2} m_2 V_{2i}'^2 \right)$$

$$\Rightarrow (e^2 - 1) \left(\frac{1}{2} m_1 V_{1i}'^2 + \frac{1}{2} m_2 V_{2i}'^2 \right) = (e^2 - 1) E_{ci}' \quad \begin{cases} e=1 & \text{ELASTICO} \\ e=0 & \text{P. ANELASTICO} \end{cases}$$

CORPO RIGIDO:

SISTEMA DI PUNTI MATERIALI CUI LA MUTUA DISTANZA RIMANE COSTANTE

- IL CENTRO DI MASSA DI UN CORPO RIGIDO HA POSIZIONE FISSATA: $\bar{r}_{cm} = \sum_i \frac{m_i \bar{r}_i}{M} \rightarrow M = \sum_i m_i$
- UN CORPO RIGIDO POSSIEME 6 GRADI DI LIBERTÀ
- IN UN CORPO RIGIDO LE FORZE INTERNE NON COMPIONO LAVORO $dS = 0$

$$\bar{r}_{cm} = \frac{1}{M} \int \bar{r} dm = \frac{\int \bar{r} dm}{\int dm}$$

$$\rho = \frac{dm}{dV}$$

DENSITÀ
VOLUMETRICA

$$\sigma = \frac{dm}{dS}$$

DENSITÀ
SUPERFICIALE

$$\lambda = \frac{dm}{dl}$$

DENSITÀ
LINEARE

$$M = \int_V \rho dV \quad \text{SE } \rho = \text{cost} \Rightarrow M = \rho \int_V dV = \rho V$$

(CORPO OMOGENEO)

$$\begin{aligned} \bar{r}_{cm} &= \frac{\int \bar{r} dm}{\int dm} = \frac{\int_V \rho \bar{r} dV}{\int_V \rho dV} \stackrel{\substack{\text{CORPO} \\ \text{OMOGENEO}}}{=} \frac{\int_V \bar{r} dV}{\int_V dV} = \\ &= \frac{1}{V} \int_V \bar{r} dV \end{aligned}$$

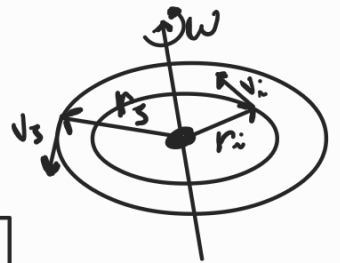
CINEMATICA DEL CORPO RIGIDO

CINEMATICA RELATIVA DI UN SDR' $\equiv$ SCH MOBILE

VELOCITÀ ANGOLARE DI ROTAZIONE DI SCH

VELOCITÀ IN SDR' $\left[\overline{V} = \underbrace{\overline{V_0'}}_{\text{VELOCITÀ DI SCH}} + \underbrace{\overline{\omega} \times \overline{r'}}_{\text{DISTANZA PUNTO DAZ CM}} = \overline{V_t} \right]$ VELOCITÀ DI TRASCINAMENTO

- TRASLAZIONE $\overline{\omega} = 0$; $\overline{V} = \overline{V_0'} = \overline{V_t}$
- ROTAZIONE $\overline{V_0'} = 0$: $\overline{V} = \overline{V_t} = \overline{\omega} \times \overline{r'}$
PER OGNI PUNTO



$\Rightarrow$ MODULO DELLA VELOCITÀ $V_i = r_i \omega$

ENERGIA CINETICA:

$$E_c = \sum_i E_{c,i} = \sum_i \frac{1}{2} m_i V_i^2 = \sum_i \frac{1}{2} m_i r_i^2 \omega^2$$

$$\Rightarrow E_c = \frac{1}{2} \underbrace{\left(\sum_i m_i r_i^2 \right)}_{I_z} \omega^2$$

MOMENTO D'INERZIA RISPETTO L'ASSE Z

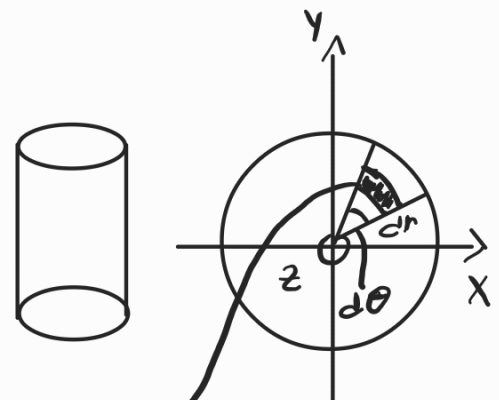
$$I_z = \int r_i^2 dm \quad dm = \rho dv$$

$$I_z = \int_V r^2 \rho dv = \rho \int_V r^2 dv$$

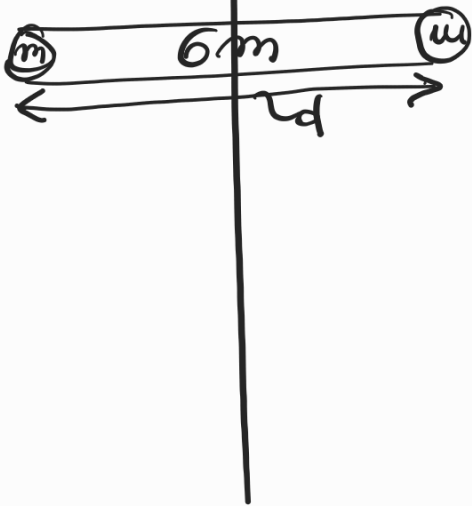
$$I_{CM} = \iiint \rho r^2 \cdot r dr d\theta dz$$

$ds = dr \cdot r d\theta$

$dv = ds dz = r dr d\theta dz$



$$\vec{\omega} = \alpha t \hat{u}_z$$



Si consideri un rotatore rigido composto:

- m puntiformi agli estremi
- $M = 6m$ (ASTA RIGIDA) $I_{\text{astro}} =$
- $\vec{\omega} = \alpha t \hat{u}_z$ $\left(\alpha \left[\frac{\pi \omega}{s^2} \right] 4 \text{ ees/ez} \cdot 2 \pi \text{ volte} \right)$

Si determini:

- Il momento $\tau_{\text{torc}} = ?$ delle forze che fanno accelerare il rotatore rigido
- Si dimostri che il Lavoro che viene fatto $\forall$ giro è dato da

$$\mathcal{L} = 2\pi \tau_{\text{torc}}$$

PROPRIETÀ $\int$ (SISTEMA CENTRO DI MASSA)

I) $\overline{r_{cm}} = 0$

DIM

$$\begin{aligned} \overline{r_i'} &= \overline{r_i} - \overline{r_{cm}} \Rightarrow \sum_i m_i \overline{r_i'} = \sum_i m_i (\overline{r_i} - \overline{r_{cm}}) = \\ &= \underbrace{\sum_i m_i \overline{r_i}}_{M \overline{r_{cm}}} - \left(\sum_i m_i \right) \overline{r_{cm}} = 0 \end{aligned} \quad \square$$

II) $\overline{v_{cm}} = 0$

DIM

$$\begin{aligned} \overline{v_i'} &= \overline{v_i} - \overline{v_{cm}} \Rightarrow \sum_i m_i \overline{v_i'} = \sum_i m_i (\overline{v_i} - \overline{v_{cm}}) = \\ &= \underbrace{\sum_i m_i \overline{v_i}}_{M \overline{v_{cm}}} - \left(\sum_i m_i \right) \overline{v_{cm}} = 0 \end{aligned} \quad \square$$

III) $M \overline{a_{cm}} = 0$

DIM

$$\left(\overline{F_i^{ext}} + \overline{F_i^{int}} \right) - \underbrace{m_i \overline{a_{cm}}}_{M \overline{a_{cm}}} = \underbrace{m_i \overline{a_i'}}_{M \overline{a_{cm}'}}$$

$$\Rightarrow \sum_i \left(\overline{F_i^{ext}} + \overline{F_i^{int}} - m_i \overline{a_{cm}} \right) = \sum_i m_i \overline{a_i'}$$

$$\Rightarrow \overline{F_{ext}} - \overline{F_{ext}} = 0 = M \overline{a_{cm}}$$

$\square$

$$IV) \boxed{\frac{d\bar{L}'}{dt} = \overline{M_{ext}}}$$

DM

$$\bar{L}' = \bar{r}' \times m_i \bar{v}' \quad \bar{L}' = \sum_i \bar{L}'_i = \sum_i \bar{r}'_i \times m_i \bar{v}'_i$$

$$\begin{aligned} \frac{d\bar{L}'}{dt} &= \frac{d}{dt} \left(\sum_i \bar{r}'_i \times m_i \bar{v}'_i \right) = \sum_i \left(\underbrace{\frac{d\bar{r}'_i}{dt}}_{\bar{v}'_i} \times m_i \bar{v}'_i \right) + \\ &+ \sum_i \left(\bar{r}'_i \times m_i \frac{d\bar{v}'_i}{dt} \right) = \sum_i \bar{r}'_i \times \underbrace{\overline{F_{i,ext}} + \overline{F_{i,int}}}_{\overline{F_{i,ext}}} + \end{aligned}$$

$$+ \sum_i m_i \bar{r}'_i \times \overline{F_{i,int}} = \overline{M_{ext}} \quad \square$$

Thermodynamische Zustandsgrößen

$$dW_i = \bar{F}_i \circ d\bar{r}_i = \left(\overline{F_{i,ext}} + \overline{F_{i,int}} \right) \circ d\bar{r}_i$$

$$dW = \sum_i dW_i = \sum_i \overline{F_{i,ext}} \circ d\bar{r}_i + \sum_i \overline{F_{i,int}} \circ d\bar{r}_i$$

$$W = W_{ext} + W_{int} = \Delta K \quad \bar{F}_{si} \circ (d\bar{r}_i - d\bar{r}_s)$$

$$dW_i = \bar{F}_i \circ d\bar{r}_i = m_i \bar{a}_i \circ d\bar{r}_i = m_i \frac{d\bar{v}_i}{dt} \circ \bar{v}_i dt = \frac{1}{2} m_i d(v_i^2)$$

$$dW = \sum_i \frac{1}{2} m_i d(v_i^2) \Rightarrow W = \int \sum_i \frac{1}{2} m_i d(v_i^2) = \sum_i \int \frac{1}{2} m_i d(v_i^2)$$

$$c) W = \sum_i \Delta K_i = \Delta K = K_f - K_i$$

$$S1 \quad \begin{cases} \text{I re coord.} & \frac{d\overline{P_{TOT}}}{dt} = \overline{P_{EXT}} \\ \text{II re coord.} & \frac{d\overline{L_{TOT}}}{dt} = -\overline{V_0} \times \overline{P_{TOT}} + \overline{M_{TOT} \times \overline{V_0}} \end{cases}$$

$$SCM \quad \begin{cases} \overline{P_{cm}} = 0 \\ \frac{d\overline{L'}}{dt} = \overline{M_{EXT}}$$

$$I) \quad \overline{L} = \overline{L'} + \overline{r_{cm}} \times \overline{P_{cm}} \quad \overline{L_{cm}}$$

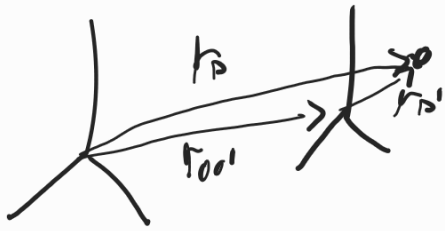
sum

$$\begin{aligned} \overline{L} &= \sum_i \overline{r_i} \times m_i \overline{v_i} = \sum_i (\overline{r_i'} + \overline{r_{cm}}) \times m_i (\overline{v_i'} + \overline{v_{cm}}) = \\ &= \underbrace{\sum_i \overline{r_i'} \times m_i \overline{v_i'}}_{\downarrow} + \underbrace{\sum_i \overline{r_i'} \times m_i \overline{v_{cm}}}_{\substack{\uparrow \overline{r_{cm}} = 0 \\ M \overline{v_{cm}} = 0}} + \underbrace{\sum_i \overline{r_{cm}} \times m_i \overline{v_i'}}_{\substack{\uparrow \overline{r_{cm}} = 0 \\ M \overline{v_{cm}} = 0}} + \underbrace{\sum_i \overline{r_{cm}} \times m_i \overline{v_{cm}}}_{\substack{\uparrow \overline{r_{cm}} = 0 \\ M \overline{v_{cm}} = 0}} \\ &= \overline{L'} + \overline{L_{cm}} \quad \square \end{aligned}$$

$$II) \quad K = K' + K_{cm}$$

$$\begin{aligned} K &= \sum_i \frac{1}{2} m_i \overline{v_i}^2 = \sum_i \frac{1}{2} m_i (\overline{v_i} \cdot \overline{v_i}) = \sum_i \frac{1}{2} m_i (\overline{v_i'} + \overline{v_{cm}}) \cdot (\overline{v_i'} + \overline{v_{cm}}) \\ &= \underbrace{\sum_i \frac{1}{2} m_i \overline{v_i'}^2}_{\downarrow} + \underbrace{2 \sum_i \overline{r_i'} \cdot m_i \overline{v_{cm}}}_{\substack{\uparrow \overline{r_{cm}} = 0 \\ M \overline{v_{cm}} = 0}} + \underbrace{\sum_i \frac{1}{2} m_i \overline{v_{cm}}^2}_{\substack{\uparrow \overline{r_{cm}} = 0 \\ M \overline{v_{cm}} = 0}} \\ &= K' + K_{cm} \quad \square \end{aligned}$$

MOVIMENTO ROTAZIONALE



$$\vec{r}_P = \vec{r}_{P'} + \vec{r}_{O'O'}$$

$$\vec{v}_P = \vec{v}_{P'} + \vec{v}_{O'O'} + \vec{\omega} \times \vec{r}_P$$

$$v_P = \underbrace{v_{O'O'}}_{V.T.T.} + \underbrace{\vec{\omega} \times \vec{r}_P}_{V.T.R.}$$

TRASLAZIONE $\omega = 0$

$$\vec{v}_P = \vec{v}_{O'O'} = \vec{v}_{cm}$$

QUANTITÀ DI MOVIMENTO $\vec{P}_{TOT} = \sum_i \vec{p}_i = \sum_i m_i \vec{v}_i = \sum_i m_i \vec{v}_{cm} =$
 $= \vec{v}_{cm} \sum_i m_i = \vec{v}_{cm} m_{TOT}$

MOMENTO ANGOLARE $\vec{L}_{TOT} = \sum_i \vec{L}_i = \sum_i \vec{r}_i \times m_i \vec{v}_i =$
 $= \sum_i \vec{r}_i \times m_i \vec{v}_{cm} = \left(\sum_i \vec{r}_i m_i \right) \times \vec{v}_{cm} = \vec{r}_{cm} m_{TOT} \times \vec{v}_{cm} =$

$$= \vec{r}_{cm} \times m_{TOT} \vec{v}_{cm} = \vec{r}_{cm} \times \vec{P}_{TOT} \quad \text{D.T. NOI VUOLIAMO} \quad \vec{L}_{TOT} = \vec{L}_{TOT}' + \vec{L}_{cm}$$

ENERGIA CINETICA $E_k = \sum_i E_{ki} = \sum_i \frac{1}{2} m_i v_i^2 = \left(\sum_i m_i \right) \frac{1}{2} v_{cm}^2 =$
 $= \frac{1}{2} m_{TOT} v_{cm}^2$

LAVORO $W_e = \int_e \vec{F}_e \cdot d\vec{s} \quad \vec{F}_e = \frac{d\vec{P}_{TOT}}{dt} = \frac{d(\vec{v}_{cm} m_{TOT})}{dt} = m_{TOT} \frac{d\vec{v}_{cm}}{dt}$

MOMENTO $\vec{M}_e = \frac{d\vec{L}_{TOT}}{dt} = \frac{d(\vec{r}_{cm} \times m_{TOT} \vec{v}_{cm})}{dt} = \frac{d\vec{r}_{cm}}{dt} \times m_{TOT} \vec{v}_{cm} +$

$$+ \vec{r}_{cm} \times m_{TOT} \frac{d\vec{v}_{cm}}{dt} = \vec{r}_{cm} \times m_{TOT} \frac{d\vec{v}_{cm}}{dt}$$

$$\vec{M}_e = \vec{r} \times \vec{F}_e \quad \vec{F}_e = m_{TOT} \frac{d\vec{v}_{cm}}{dt}$$

$$\vec{M}_e = \vec{r}_{cm} \times \vec{F}_e = \vec{r}_{cm} \times m_{TOT} \frac{d\vec{v}_{cm}}{dt} = 0$$

NO ROTAZIONE SE $\vec{F}$ APPLICATA SUL CENTRO DI MASSA

ROTAZIONE

$$\vec{V}_P = \vec{V}_{OO'} + \vec{\omega} \times \vec{r}_{P'}$$

$$\vec{V}_{OO'} = 0$$

ROTAZIONE ATTORNO
AD UN ASSE FISSO

$$\vec{V}_P = \vec{\omega} \times \vec{r}_{P'} = \vec{V}_P$$

$$\vec{V}_P = \vec{\omega} \times \vec{r}_P = \omega r_P \sin \alpha \hat{u}_T \rightarrow \vec{V}_P = \omega R_P \hat{u}_T$$

MOMENTO ANGOLARE

$$\vec{L}_{TOT} = \sum_i \vec{L}_i = \sum_i \vec{r}_i \times m_i \vec{V}_i = \sum_i (\vec{r}_i + \vec{R}_i) \times m_i \omega R_i \hat{u}_T$$

$$= \sum_i (\vec{r}_i + \vec{R}_i) \times m_i \omega R_i \hat{u}_T = \sum_i \vec{r}_i \times m_i \omega R_i \hat{u}_T + \sum_i \vec{R}_i \times m_i \omega R_i \hat{u}_T =$$

$$① = \sum_i m_i \omega R_i \underbrace{\vec{r}_i \times \hat{u}_T}_{-\vec{r}_i \hat{u}_T} = - \sum_i m_i \omega r_i R_i \hat{u}_T = - \sum_i m_i \omega r_i \vec{R}_i$$

$$② = \sum_i m_i \omega R_i \vec{R}_i \times \hat{u}_T = \sum_i m_i R_i^2 \underbrace{\omega \hat{u}_T}_{\vec{\omega}} = \boxed{\sum_i m_i R_i^2} \vec{\omega}$$

$$\vec{L}_{TOT} = \underbrace{- \sum_i m_i \omega r_i \vec{R}_i}_{\vec{L}_1} + \underbrace{\sum_i m_i R_i^2 \omega}_{\vec{L}_2 = I_z \vec{\omega}}$$

ENERGIA CINETICA $E_K = \sum_i \frac{1}{2} m_i \underbrace{V_i^2}_{\omega R_i^2} = \sum_i \frac{1}{2} m_i \omega^2 R_i^2 = \frac{1}{2} I_z \omega^2$

LAGRANGE $T_E = \frac{d \overline{P_{TOT}}}{dt} = m \omega \frac{d \overline{V_{TOT}}}{d\theta}$

$$\overline{M}_\theta = \frac{d \overline{L_{TOT}}}{d\theta} = \frac{d \overline{L_1}}{d\theta} + \frac{d \overline{L_2}}{d\theta}$$

$$\Rightarrow \overline{M}_\theta = \frac{d \overline{L_2}}{d\theta} = \frac{d (I_z \omega)}{d\theta} = \frac{d I_z}{d\theta} \omega + I_z \frac{d \omega}{d\theta}$$

$$\text{Lavoro } E_k = \frac{1}{2} I_2 \omega^2 \Rightarrow \frac{dE_k}{d\omega} = I_2 \omega$$

$$\begin{aligned} W_e &= \Delta E_k \Rightarrow dW_e = dE_k = I_2 \omega d\omega = \\ &= I_2 \frac{d\theta}{dt} d\omega = I_2 \bar{\omega} d\theta = M_2 d\theta \end{aligned}$$

$$W_e = \int_{\theta_1}^{\theta_2} M_2 d\theta$$

TRASLAZIONE

$$\overline{P_{tot}} = m_{tot} \overline{V_{cm}}$$

$$E_k = \frac{1}{2} m_{tot} V_{cm}^2$$

$$\overline{E} = m_{tot} \overline{v_{cm}}$$

ROTAZIONE

$$\overline{L_2} = I_2 \overline{\omega}$$

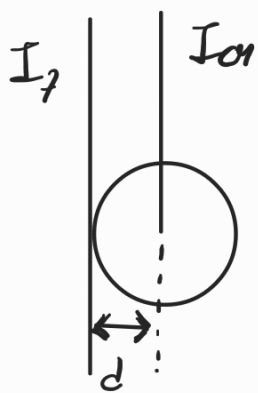
$$E_k = \frac{1}{2} I_2 \omega^2$$

$$\overline{M_2} = I_2 \overline{\alpha}$$

MOMENTO D'INERZIA

$$I_{cm} = \sum_i m_i R_i^2$$

$$I = \int_{tot} r^2 dm = \int_{tot} r^2 \rho dV$$



TEOREMA DI HUYGENS-STEINER

$$I_z = I_{CM} + M d^2 \quad I_z \parallel I_{CM}$$

DIM

$$\begin{aligned} I_z &= \sum_i m_i R_i^2 = \sum_i m_i (x_i^2 + y_i^2) = \sum_i m_i [x_i'^2 + (y_i' + d)^2] = \\ &= \sum_i m_i x_i'^2 + \sum_i m_i y_i'^2 + \sum_i m_i d^2 + \sum_i m_i 2d y_i' \\ I_z &= \sum_i m_i (x_i'^2 + y_i'^2) + d^2 \underbrace{\sum_i m_i}_{M} + 2d \underbrace{\sum_i m_i y_i'}_{y'_{CM} = 0} \\ I_z &= I_{CM} + M d^2 \end{aligned}$$

CONDIZIONE DI PURO ROTOLAMENTO

$$\overline{\mathcal{O}_{CM}} = -\overline{\omega} \times \overline{R_C}$$

$$\Rightarrow \frac{\overline{F_e}}{m} = -\frac{\overline{M_z}}{I_{CM}} \times \overline{R_C}$$

STATICA DEI FLUIDI

$$p = \frac{F}{S}$$

$$\overline{\nabla p} = \rho \overline{f_v} = 0$$

LEGGI DI STEVINO

$$p(h) = p_0 + \rho g h$$

PRINCIPIO DI ARCHIMEDE

$$\overline{F_A} = - \rho_{liq} V \overline{g} \quad \text{APPLICATA NEL BARICENTRO}$$

PRINCIPIO ZERO DELLA TERMODINAMICA

DUE SISTEMI A E B in INDIPENDENZA LENTAMENTE in EQUILIBRIO TERMICO con un SISTEMA C sono in EQUILIBRIO TERMICO

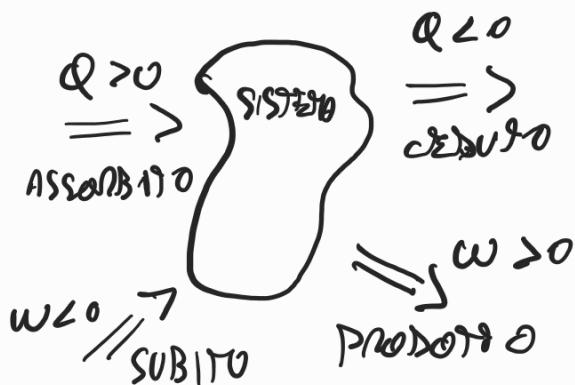
PRIMO PRINCIPIO DELLA TERMODINAMICA

$$Q - W = \Delta U$$

CALORIMETRIA

$$Q = \int c(T) m dT$$

$$dQ = c m dT$$



CALORE SPECIFICO

$$c = \frac{dQ}{m dT}$$

TRASFORMAZIONE REVERSIBILE

$$i \rightarrow f \rightarrow i$$

- QUASI STATICA Q.S.

- ~~SENZA~~ ATTRI (DISSIPAZIONE DI CALORE)

GAS IDEALI

$$PV = nRT$$

ENERGIA INTERNA

$$dU = n c_v dT$$

TRASFORMAZIONE ADIABATICA Q.S.

$$TV^{\gamma-1} = \text{cost}_1$$

$$PV^{\gamma} = \text{cost}_2$$

$$TP^{\frac{1-\gamma}{\gamma}} = \text{cost}_3$$

$$\gamma = \frac{C_p}{C_v}$$

TRASFORMAZIONE ISOTERMA

$$PV = \text{cost}$$

TRASFORMAZIONE POLITROPICA

$$PV^{\alpha} = \text{cost}$$

$$\alpha = \frac{C_p - C}{C_v - C}$$

SECONDO PRINCIPIO DELLA TERMODINAMICA

$\eta > 1$: NO in TERM $Q > 0$ E SOLO $W > 0$

CLAUSIUS: NO in FRIGO $Q > 0$ E SOLO $W = 0$

RENDIMENTO (CICLO TERMICO)

$$\eta = \frac{W}{Q_{AS}} \quad \eta_C = 1 - \frac{T_C}{T_H}$$

$$\eta_{REV} = \eta_C \quad \eta_{IRR} < \eta_C$$

EFFICIENZA (CICLO FRIGORIFERO)

$$E = \frac{Q_{AS}}{|W|}$$

ENTROPIA

$$dS = \frac{dQ}{T} \Big|_{REV} \quad \Delta S = \int \frac{dQ}{T} \Big|_{REV}$$

$$S_B = S_A + \int_A^B \frac{dQ}{T} \Big|_{REV}$$

$$\Delta S_U = \Delta S_S + \Delta S_A$$

$$\text{- TRANS. REV.} \quad \Delta S_U = 0 \quad \Delta S_A = -\Delta S_S$$

$$\text{- TRANS. IRR} \quad \Delta S_U > 0 \quad \Delta S_A > -\Delta S_S$$

$$\text{CICLO TERMODINAMICO} \quad \Delta S_S = 0$$

$$\text{- CICLO REV} \quad \Delta S_U = 0 \quad \Delta S_A = 0$$

$$\text{- CICLO IRR} \quad \Delta S_U > 0 \quad \Delta S_A > 0$$

$$E_{IN} = W_{REV} - W_{IRR}$$

$$R(t) = \frac{(v_0^2 + g^2 t^2 - 2 v_0 y \sin \theta t)^{3/2}}{v_0 \cos \theta g}$$

$$R(t) = \frac{(v_{0x}^2 + v_{0y}^2 + g^2 t^2 - 2 v_{0y} g t)^{3/2}}{v_{0x} g}$$

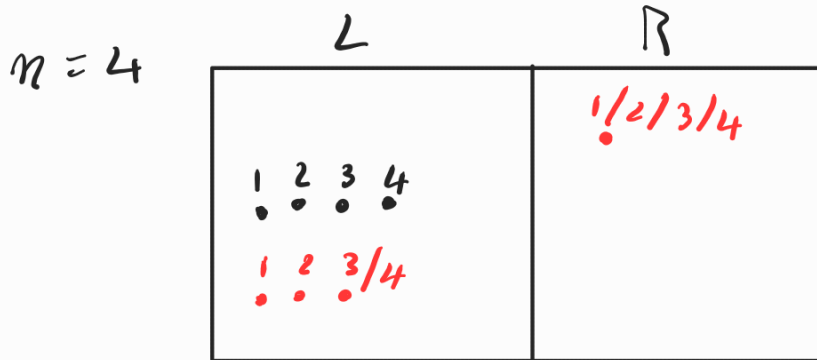
$$R(t) = \frac{(v_{0x}^2 + (v_{0y} - g t)^2)^{3/2}}{v_{0x} g}$$

$$\text{ENTROPIA } W_{\text{REV}} - W_{\text{IRR}} = T_2 \Delta S$$

INTERPRETAZIONE STATISTICA ENTROPIA

- STATO TERMODINAMICO $\rightarrow (P, V, T)$
- STATO DINAMICO $\rightarrow (x, v, z)$

TANTI STATI DINAMICI (MICROSTATI) CORRISPONDONO
ALLO STESSO STATO TERMODINAMICO (MACROSTATO)



① TUTTE LE $n = 4$
PARTICELLE A SX

$\Gamma = 1$
DI COMBINAZIONI
POSSIBILI

② 2 A SX E 2 A DX $\Gamma = 4$

③ $n_L = 2$ $n_R = 2 \Rightarrow \Gamma = 6$

MAX $\Gamma = 6$ DISORDINE MASSIMO

MIN $\Gamma = 1$ ORDINE MASSIMO

$$\Gamma = \frac{n!}{n_L! n_R!}$$

$$\textcircled{2} \Gamma = \frac{4 \cdot 3 \cdot 2 \cdot 1}{2 \cdot 2 \cdot 1 \cdot 1} = 4$$

100 PARTICELLE $\Rightarrow \Gamma = \frac{8 \cdot 10^{157}}{8 \cdot 10^{64} \cdot 8 \cdot 10^{64}} = 10^{33}$

$$S = k_B \ln(\Gamma)$$

EQUAZIONE DI BOLZMANN

↓
COSTANTE DI BOLZMANN $\frac{R}{N_A}$

TEORIA CINETICA DEI GAS IDEALI

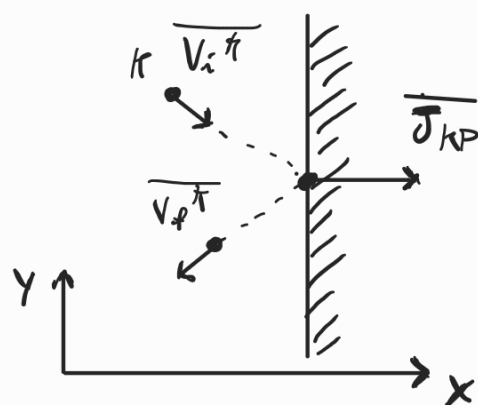
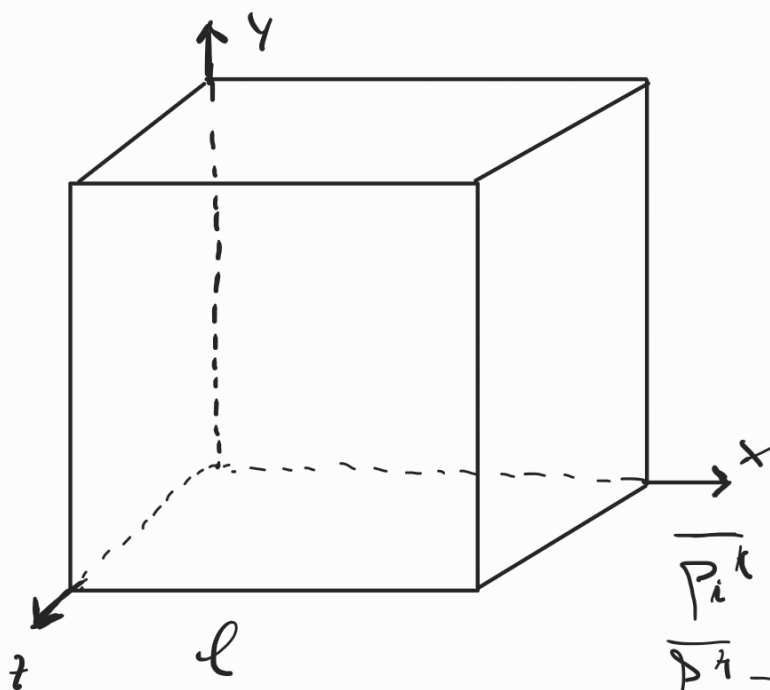
IPOTESI:

- 1) MOLECOLE NON INTERAGENTI $\Rightarrow \overline{A} = \overline{F} \Rightarrow E_p = \text{cost} = 0$
 $\hookrightarrow$ SI MUOVONO A COSTO COSTANTE UNIFORME
- 2) VELOCITÀ DELLE MOLECOLE NON HA UNA DIREZIONE PREFERENZIALE $\Rightarrow$ NON CASUALE
- 3) URTI TRA MOLECOLE O TRA MOLECOLE E PARETI SONO ISTANTANEI ED ELASTICI
- 4) VOLUME DELLE MOLECOLE MOLTO PIU' PICCOLO DEL VOLUME DEL CONTENITORE $\Rightarrow V_{\text{CONTENITORE}} \gg V_{\text{MOLECOLE}}$
 $\Rightarrow$ MOLECOLE PUNTIFORMI

$\Downarrow$

$P \ll P_{\text{atm}} \rightarrow$ GAS RAREFATTO

$T > T_c \rightarrow$ GAS CHE NON SUBISCE TRANSIZIONI DI FASE

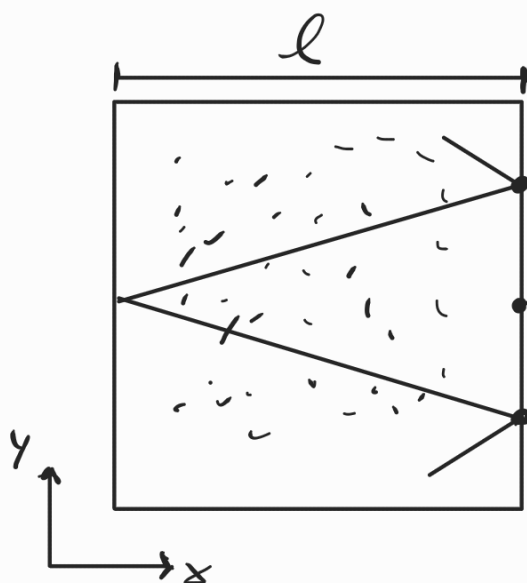


$$\overline{p}_i^k = m_k \overline{v_i^k} = m_k \left(v_x^k \hat{u}_x + v_y^k \hat{u}_y + v_z^k \hat{u}_z \right)$$

$$\overline{p}_f^k = m_k \overline{v_f^k} = m_k \left(-v_x^k \hat{u}_x + v_y^k \hat{u}_y + v_z^k \hat{u}_z \right)$$

$$\Delta \overline{p}_k = \overline{p}_f^k - \overline{p}_i^k = -2 m_k v_x^k \hat{u}_x \quad \overline{J}_{p_k} = \Delta \overline{p}_k = -2 m_k v_x^k \hat{u}_x$$

3° PRINCIPIO DINAMICA $\Rightarrow \overline{J_{KP}} = -\overline{J_{PK}} = 2 m_H v_x^* u_x^1$



$$t_1 = \frac{2l}{v_x^*}$$

TEMPO TRA DUE URTI CON PARETE SX

$$\overline{F_{\perp}^x} \Delta t \rightarrow N_U = \frac{\Delta t}{t_1} = \frac{\Delta t}{2l} v_x^*$$

INTERVALLO DI TEMPO CASUALE NUMERO URTI IN Δt

$$\overline{J_{tot, k}} = N_U \cdot J_{KD} = \Delta t \frac{m_H}{2l} v_x^{*2} 2$$

IMPULSO TOTALE DI K NEL Δt

IMPULSO TOTALE DI TUTTE LE MOLECOLE IN Δt

$$J_{tot} = \sum_k \overline{J_{tot, k}} = \sum_k \Delta t \frac{m_k}{l} v_x^{*2} =$$

$$J_{tot} = \Delta t \frac{m}{l} \left[\sum_k v_x^{*2} \right]$$

$\Downarrow$

$$\langle v_x^2 \rangle = \frac{1}{N} \sum_k (v_x^*)^2$$

VELOCITÀ QUADRATICA MEDIA LUNGO X

$$\Rightarrow J_{tot} = \Delta t \frac{m}{l} N \langle v_x^2 \rangle$$

$$\overline{F_{\perp}^x} = \frac{J_{tot}}{\Delta t} = \frac{m}{l} N \langle v_x^2 \rangle$$

$$P_x = \frac{F_{\perp}^x}{S} = \frac{m}{\underbrace{l \cdot S}_V} N \langle v_x^2 \rangle = \frac{m}{V} N \langle v_x^2 \rangle$$

$$P_y = \frac{m}{V} N \langle v_y^2 \rangle \quad P_z = \frac{m}{V} N \langle v_z^2 \rangle$$

$$P = \frac{m}{V} N \langle v_x^2 \rangle$$

$$\langle v_x^2 \rangle = \langle v_y^2 \rangle = \langle v_z^2 \rangle$$

$$V = \sqrt{v_x^2 + v_y^2 + v_z^2}$$

$$\begin{aligned} \langle v^2 \rangle &= \langle v_x^2 \rangle + \langle v_y^2 \rangle + \langle v_z^2 \rangle = \\ &= 3 \langle v_x^2 \rangle \end{aligned}$$

$$\Rightarrow P = \frac{m}{V} N \frac{1}{3} \langle v^2 \rangle$$

$$\Rightarrow P = \frac{2m}{2V} N \frac{1}{3} \langle v^2 \rangle$$

$$\Rightarrow P = \frac{2N}{3V} \underbrace{\frac{1}{2} m \langle v^2 \rangle}_{\langle E_k \rangle} \Rightarrow$$

$$P_{\text{gas}} = \frac{2}{3} \frac{N}{V} \langle E_k \rangle$$

EQUAZIONE DI

STATE - CLAUSIUS - KÖNIG

$$\left. \begin{array}{l} \text{TEORIA CINETICA} \Rightarrow PV = \frac{2}{3} N \langle E_k \rangle \\ \text{SPERIMENTALE} \Rightarrow PV = nRT \end{array} \right\}$$

$$\Rightarrow \frac{2}{3} N \langle E_k \rangle = \underbrace{n}_{\frac{N}{N_A}} RT = \frac{N}{N_A} \overset{k_B}{\underbrace{R}} T$$

$$\Rightarrow \frac{2}{3} \langle E_k \rangle = k_B T \Rightarrow$$

$$T = \frac{2 \langle E_k \rangle}{3 k_B}$$

$$\Leftrightarrow \langle E_k \rangle = \frac{3}{2} k_B T$$

$$\frac{1}{2} m \langle v^2 \rangle = \frac{3}{2} k_B T \Rightarrow 3 \left(\frac{1}{2} m \langle v_x^2 \rangle \right) = 3 \frac{1}{2} k_B T$$

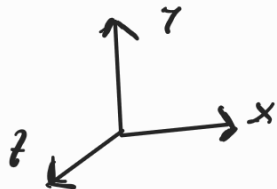
$$\Rightarrow \langle E_{k,x} \rangle = \frac{1}{2} k_B T$$

TEOREMA DI EQUIPARTIZIONE
DELL'ENERGIA

IN UN SISTEMA CON g GRADI DI LIBERTÀ OGNI GRADO
CONTRIBUISCE ALL'ENERGIA MEDIA DEL SISTEMA CON $\frac{1}{2} k_B T$

GAS MONOATOMICO

$$g = 3$$



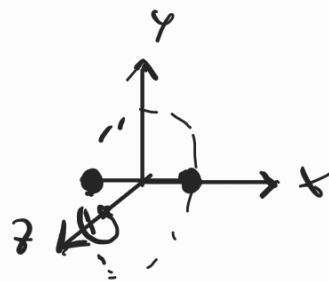
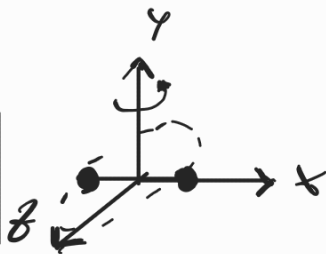
$$\langle E_k \rangle = \frac{g}{2} k_B T = \frac{3}{2} k_B T$$

GAS BIATOMIC

$$g_T = 3$$

$$g_R = ?$$

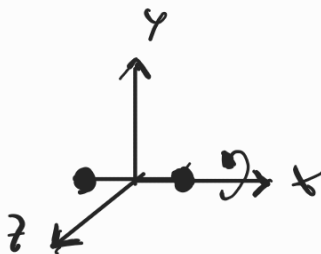
$$g = 5$$



ROTATIONI

$$E_{rot y} = \frac{1}{2} I_y \omega_y^2$$

$$E_{rot x} = \frac{1}{2} I_x \omega_x^2 = 0$$



ENERGIA INTERNA

$$U = \sum_i (E_{k,i} + \underbrace{E_{v,i}}_{\text{IDOTTI 1}}) \Rightarrow U = \sum_i E_{k,i} = N \langle E_k \rangle$$

IDOTTI 1
TRADE CINETICA

$$\Rightarrow U = N \frac{g}{2} k_B T$$

CALORE SPECIFICO

$$C = \frac{1}{n} \frac{dQ}{dT} \Big|_{\text{TRASF.}} \rightarrow C_V = \frac{1}{n} \frac{dQ}{dT} \Big|_{V=\text{cost}}$$

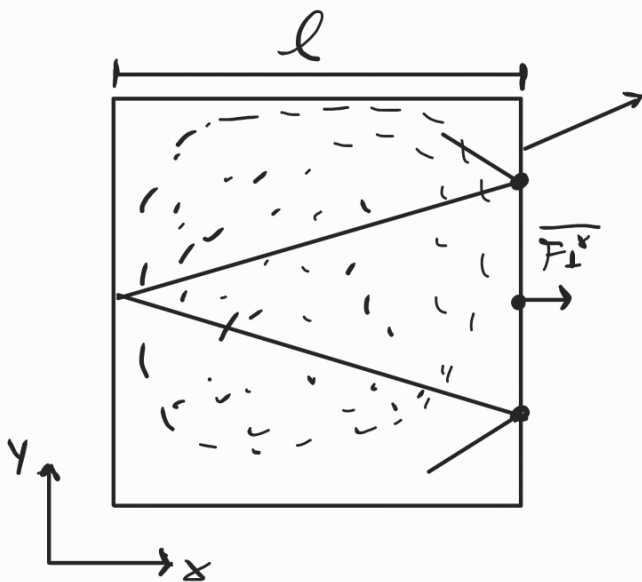
$$\Rightarrow \underset{Q=U}{W=0} \Rightarrow C_V = \frac{1}{n} \frac{1}{dT} \left(U \frac{g}{2} k_B T \right)$$

$$\Rightarrow C_V = \frac{1}{\underbrace{n}_{\frac{N}{N_A}}} \frac{g}{2} k_B \Rightarrow C_V = \frac{g}{2} \underbrace{(k_B N_A)}_R$$

$$C_V = \frac{g}{2} R$$

$$\rightarrow \text{MONOATOMICO } g=3 \quad C_V = \frac{3}{2} R$$

$$\rightarrow \text{BIATOMICO } g=5 \quad C_V = \frac{5}{2} R$$



STATISTICAMENTE DATO L'ELEVATO
NUMERO DI PARTICELLE MEDIANTE
E COME SE NON CI FOSSE UNO
(QUASI SICURAMENTE CI SARÀ UN'ALTRA
MOLECOLA CHE SEGUIRÀ LA TRAIETTORIA
DI UNA MOLECOLA CHE HA URTO)

$$\psi(v) = \text{cost} \cdot v^2 \cdot e^{-\text{cost} \cdot v^2}$$

DISTRIBUZIONE DI
MAXWELL - BOLZMANN

